

1 AssociationExplorer: A user-friendly Shiny application 2 for exploring associations and visual patterns

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10 Abstract

AssociationExplorer is an open-source interactive R Shiny application designed to help non-technical users explore statistical associations within multivariate datasets. Aimed particularly at journalists, educators, and engaged citizens, the tool facilitates the discovery and interpretation of meaningful patterns between variables without requiring programming or statistical expertise. Users can upload structured data (e.g., from surveys or open government datasets), select relevant variables, and dynamically visualize relationships via a correlation network and contextual bivariate plots. To illustrate its capabilities, we present a case study based on the European Social Survey (ESS), showcasing how users can investigate links between attitudes, behaviors, and socio-demographic indicators across countries. The app supports a range of association measures adapted to variable types (Pearson's r , Eta, and Cramer's V), ensuring both flexibility and statistical rigor. The visual interface enables users to adjust thresholds for association strength and examine results through interactive graphs and summary tables, making the app particularly well-suited for data storytelling, exploratory research, and public communication. AssociationExplorer demonstrates how open-source statistical tooling can enhance transparency, accessibility, and insight in the interpretation of complex social data.

11 **Keywords:** R Shiny, Exploratory data analysis, Correlation network

12 Metadata

13 The metadata associated with the current version of the software is summa-
14 rized in Table 1. The application has been developed and tested using R

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version 4.4.1 [15], ensuring compatibility with the package versions listed in the repository documentation.

Nr.	Code metadata description	Metadata
C1	Current code version	v3.5.7
C2	Permanent link to code/repository used for this code version	https://github.com/AntoineSoetewey/AssociationExplorer
C3	Permanent link to Reproducible Capsule	https://codeocean.com/capsule/7707308/tree
C4	Legal Code License	MIT License
C5	Code versioning system used	Git/GitHub
C6	Software code languages, tools, and services used	R, R Shiny
C7	Compilation requirements, operating environments & dependencies	See a list of the required R packages at https://github.com/AntoineSoetewey/AssociationExplorer/blob/main/packages.md
C8	If available link to developer documentation/manual/screencast	https://github.com/AntoineSoetewey/AssociationExplorer/tree/main/documentation
C9	Support for questions or issues	https://github.com/AntoineSoetewey/AssociationExplorer/issues

Table 1: Code metadata

The remainder of this paper is structured as follows. Section 1 outlines the motivation and significance of the application, highlighting its contribution to data accessibility and exploratory analysis. Section 2 provides a detailed description of the software’s architecture and functionalities. Section 3 presents an illustrative example using data from the European Social Survey. Section 4 discusses the potential impact of the application in research, education, and journalism. Finally, Section 5 concludes with a summary and future directions.

1. Motivation and significance

The growing availability of large, complex, and high-dimensional datasets in the social sciences and public policy domains offers unprecedented opportu-

28 nities for insight but also presents significant challenges for exploration and
29 interpretation, particularly for non-specialist audiences. Journalists, educa-
30 tors, and engaged citizens often struggle to identify and interpret meaningful
31 relationships between variables without the aid of programming skills or for-
32 mal statistical training. This barrier limits the broader societal impact of
33 open data initiatives, which are designed to promote transparency, account-
34 ability, and informed public discourse.

35 To address this gap, we developed AssociationExplorer, a free, open-source
36 R Shiny [1] application that enables intuitive and statistically grounded ex-
37 ploration of multivariate associations. The tool guides users through a visual
38 journey of variable relationships by automatically computing appropriate bi-
39 variate association measures—Pearson’s r , Eta, and Cramer’s V —depending
40 on variable types, and presenting the results in an interactive correlation
41 network. Users can set thresholds for the strength of association and explore
42 linked bivariate plots or tables with descriptive labels. This workflow sup-
43 ports transparent, reproducible, and non-technical exploratory data analysis
44 (EDA).

45 Our software is particularly suited to survey-based datasets and public opin-
46 ion studies. As an illustrative case, we apply AssociationExplorer to the
47 European Social Survey (ESS), a cross-national survey that collects attitudi-
48 nal, behavioral, and socio-demographic data across European countries. The
49 tool allows users to uncover associations between trust in institutions, pol-
50 icy preferences, media usage, and demographic characteristics without any
51 coding. This type of interactive analysis can empower journalists to build
52 data-driven narratives, educators to teach statistical thinking, and citizens
53 to explore evidence underlying public debate.

54 While several tools and libraries exist for correlation analysis (e.g., `corrr`
55 [8], `GGally` [18], `corrplot` [22], `ggstatsplot` [14], `correlation` [11, 12],
56 `lares` [9] and `Hmisc` [7] in R [15], or Python packages like `seaborn` [21]
57 and `pingouin` [20]), they typically require programming proficiency and fo-
58 cus primarily on numerical associations. Most of these tools do not handle
59 nominal categorical variables directly; if included, such variables are often
60 transformed using one-hot or dummy encoding, which can transform their
61 original structure and limit interpretations.

62 In contrast, AssociationExplorer is designed to handle both quantitative and
63 qualitative variables (including nominal factors) natively and transparently.
64 It provides a guided, end-to-end workflow that begins with data upload and
65 preprocessing, continues through variable selection and association filtering,
66 and ends with interpretable visualizations. This structured process is in-
67 tuitive and accessible for users of all backgrounds, making the app espe-
68 cially suitable for those without programming experience or formal statistical

69 training. By lowering the technical barrier for statistical exploration, Asso-
70 ciationExplorer contributes to a more inclusive data culture and supports
71 data-driven discovery in both academic and public-facing contexts.

72 2. Software description

73 2.1. Software architecture

74 Upon upload, the dataset is preprocessed to exclude variables with zero vari-
75 ance, as these variables do not vary across observations and therefore cannot
76 contribute to meaningful associations or visualizations. Removing them helps
77 reduce noise and ensures that only informative variables are included in the
78 analysis. Optionally, the user can provide a variable description file, which
79 is integrated and used to annotate visual elements. The backend computes
80 association measures tailored to the variable types: Pearson's r for numeric
81 pairs, Cramer's V for categorical pairs, and the correlation ratio (η) for mixed
82 pairs. Associations are filtered using user-defined thresholds and represented
83 in a correlation network and complementary bivariate plots. The app handles
84 both CSV and Excel files.

85 The application is currently available as a standalone open-source R Shiny
86 application on GitHub.¹ It can be launched directly from an R session using
87 the following command, provided the `shiny` package is installed:

```
88 library(shiny)  
89 runGitHub("AssociationExplorer", "AntoineSoetewey")
```

90 A fully web-based version, requiring no installation or technical setup, is also
91 planned as part of its integration into a broader online platform developed
92 within the ODALON (Open multimodal Data for Automated Local News)
93 research project, which aims to support the production of local news (more
94 information about this project in Sections 3 and 4).

95 2.2. Software functionalities

96 The major functionalities of the AssociationExplorer application include:

- 97 • **Data upload and cleaning:** The app supports CSV and Excel files.
98 It automatically removes variables with only one unique value, as they
99 lack variability and cannot contribute to association analyses. Addi-
100 tionally, it can optionally integrate user-supplied descriptions of vari-
101 ables, which are used to enhance the clarity and interpretability of

¹<https://github.com/AntoineSoetewey/AssociationExplorer>

visualizations, particularly for non-technical users. The description file must be structured with two columns: **Variable** (containing the exact variable names as in the dataset) and **Description** (containing the corresponding descriptions).

- **Variable selection interface:** Users can interactively choose which variables to explore. When a description file is provided, a summary table links variable names to their descriptions.
- **Dynamic association filtering:** The app computes pairwise association measures between all selected variables, using a method tailored to the types of variables involved:

- For pairs of numeric variables X and Y , the app calculates Pearson’s correlation coefficient (r), and retains the association if the coefficient of determination (R^2) exceeds a user-defined threshold:

$$R^2 = r^2 = (\text{cor}(X, Y))^2 \quad (1)$$

where the Pearson’s correlation coefficient $\text{cor}(X, Y)$ is defined as:

$$r(X, Y) = \frac{\sum_{i=1}^n (X_i - \bar{X})(Y_i - \bar{Y})}{\sqrt{\sum_{i=1}^n (X_i - \bar{X})^2} \sqrt{\sum_{i=1}^n (Y_i - \bar{Y})^2}} \quad (2)$$

where $\bar{X}$ and $\bar{Y}$ are the sample means of X and Y , respectively, and n is the number of observations.

- For pairs of categorical variables, it computes Cramer’s V, a normalized measure of association derived from the chi-squared statistic:

$$V = \sqrt{\frac{\chi^2}{n \cdot \min(k - 1, r - 1)}} \quad (3)$$

where χ^2 is the chi-squared statistic, n is the total number of observations, and k, r are the numbers of categories in each variable.

- For mixed pairs (one numeric and one categorical variable), the app computes the correlation ratio (η), which quantifies how much of the variance in the numeric variable is explained by the grouping structure of the categorical variable. It is defined as:

$$\eta = \sqrt{\frac{SS_{\text{between}}}{SS_{\text{total}}}} \quad (4)$$

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where:

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- SS_{total} is the *total sum of squares* of the numeric variable:

$$SS_{\text{total}} = \sum_{i=1}^n (y_i - \bar{y})^2$$

129

with y_i the observed numeric values and $\bar{y}$ their overall mean.

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- SS_{between} is the *between-group sum of squares*, computed as:

$$SS_{\text{between}} = \sum_{g=1}^G n_g (\bar{y}_g - \bar{y})^2$$

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where G is the number of groups (categories), n_g is the number of observations in group g , $\bar{y}_g$ is the group mean, and $\bar{y}$ is the overall mean.

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This formulation captures the proportion of the total variance in the numeric variable that can be attributed to differences between the categorical groups. A pair is retained only if η^2 exceeds the numeric threshold defined by the user.

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Each association is retained only if its corresponding strength metric— R^2 , η^2 , or Cramer’s V —exceeds the threshold set by the user. These thresholds can be adjusted interactively through the interface, and the filtering process is reactive: updates to the thresholds immediately propagate to the network and bivariate visualizations. This allows users to dynamically control the sensitivity of the association analysis and focus on relationships of substantive interest.

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- **Interactive correlation network:** The filtered associations are displayed as an interactive graph where nodes represent variables and edges represent associations. Edge thickness and length reflect the strength of the association: stronger associations are shown with thicker and shorter edges, whereas weaker associations are displayed with thinner and longer edges. For quantitative–quantitative pairs specifically, the color of the edges conveys direction: red edges indicate negative associations, while blue edges indicate positive ones. This combination of visual cues helps users quickly identify the most meaningful relationships in the network. When hovering over nodes, variable descriptions are displayed if a description file has been provided by the user; otherwise, the variable names are shown by default. This allows for

quick access to additional context without cluttering the visualization, enhancing interpretability while keeping the network clean and readable. The network is built using the `visNetwork` R package, which supports interactive and customizable graph layouts. Full documentation is available at <https://datastorm-open.github.io/visNetwork/>.

- **Bivariate visualization of variable pairs:** For each variable pair exceeding the threshold:

- Scatter plots with linear regression lines are shown for numeric pairs, helping visualize the direction and strength of the relationship. The regression line is computed using the ordinary least squares (OLS) method, which minimizes the sum of squared vertical distances between the observed data points and the fitted line.
- Colored contingency tables with marginal sums are shown for categorical pairs, where cell background colors vary in intensity according to the frequency of observations, using a blue gradient to highlight higher counts.
- Mean plots are shown for numeric-categorical pairs, with bars ordered by mean value to make it easy to compare and rank categories based on the quantitative variable.

Confidence intervals for the regression lines and standard errors in the mean plots are intentionally omitted to maintain a clean, uncluttered visualization that prioritizes ease of interpretation. Mean plots were selected over boxplots to avoid overwhelming non-expert users with distributional information, focusing instead on clear, accessible insights about average group differences.

- **Accessibility and user guidance:** A dedicated help section explains each step, allowing users with a limited statistical background to interactively explore their data.

2.3. Sample code snippets analysis

Below is a representative snippet from the application showing how the software selects the appropriate association measure depending on the types of the variable pair and filters associations based on user thresholds:

```
# Numeric vs numeric case
if (is_num1 && is_num2) {
  ...
}
```

```

193   r <- cor(x, y, use = "complete.obs")
194   cor_val <- ifelse(r^2 >= threshold_num, r, 0)
195   cor_type <- "Pearson's r"
196
197   # Categorical vs categorical case
198 } else if (!is_num1 && !is_num2) {
199   ...
200   tbl <- table(x, y)
201   ...
202   n_obs <- sum(tbl)
203   df_min <- min(nrow(tbl) - 1, ncol(tbl) - 1)
204   if (df_min > 0) {
205     v_cramer <- sqrt(chi$statistic / (n_obs * df_min))
206     cor_val <- ifelse(v_cramer >= threshold_cat,
207                       v_cramer, 0)
208     cor_type <- "Cramer's V"
209   }
210
211   # Mixed case (numeric vs categorical)
212 } else {
213   ...
214   means_by_group <- tapply(num_var, cat_var,
215                             mean, na.rm = TRUE)
216   overall_mean <- mean(num_var, na.rm = TRUE)
217   n_groups <- tapply(num_var, cat_var, length)
218   bss <- sum(n_groups * (means_by_group - overall_mean)^2,
219             na.rm = TRUE)
220   tss <- sum((num_var - overall_mean)^2, na.rm = TRUE)
221
222   if (tss > 0) {
223     eta <- sqrt(bss / tss)
224     cor_val <- ifelse(eta^2 >= threshold_num, eta, 0)
225     cor_type <- "Eta"
226   }
227 }

```

228 This conditional structure ensures that the correct statistical method is ap-
229 plied for each type of variable pair, supporting a robust and interpretable
230 exploration of associations.

3. Illustrative example

To demonstrate the core functionalities of AssociationExplorer, we use a curated subset of data from the European Social Survey (ESS), Round 11. The ESS is a large-scale, cross-national survey that measures attitudes, beliefs, and behaviors across European countries. The original dataset includes responses from over 46,000 individuals on topics such as politics, trust, well-being, media use, and health. The full ESS dataset, codebook, and documentation are freely available at <https://ess.sikt.no/en/> [3, 2].

For this example, we focus on the Belgian respondents, resulting in a reduced dataset of 1,594 individuals. We selected 60 variables covering areas highly relevant for understanding public opinion and everyday life in Belgium: interest in politics, confidence in institutions, lifestyle behaviors, perceived discrimination, vaccination, and more. These variables include both numbers (quantitative data) and labels or categories (qualitative data), making the dataset ideal for exploring diverse forms of associations.

This example is particularly relevant for our research project ODALON, which aims to develop a platform that supports the (semi-)automated production of local news in Belgium. AssociationExplorer plays a key role in this effort by offering journalists, researchers, and citizens an intuitive tool to explore potentially newsworthy patterns in public and survey data, without requiring programming skills or statistical training.

Data preparation for the curated dataset was carried out in R and included:

- Filtering the dataset to include only Belgian respondents.
- Converting survey-specific nonresponse codes (e.g., 77, 88, 9999, etc.) to NA values, based on the ESS codebook.
- Recoding several categorical variables to have meaningful and interpretable labels (e.g., for gender, religion, political participation, or health behaviors).

The full R script used to perform this transformation is openly available in the `data` folder of the GitHub repository at <https://github.com/AntoineSoetewey/AssociationExplorer/tree/main/data>.

Once the dataset is uploaded into AssociationExplorer via the `Data` tab (see Figure 1), users are guided through a step-by-step process. In addition to the main dataset, users can optionally upload a separate description file that provides human-readable explanations for each variable. In our example, this file was created using information from the official ESS codebook, allowing for clearer interpretation throughout the app interface. If no description

Association Explorer

Upload your dataset (CSV or Excel)

Browse... ESS11_BE_data.csv
Upload complete

(Optional) Upload variable descriptions (CSV or Excel)

Browse... ESS11_BE_description.csv
Upload complete

The descriptions file must contain exactly two columns named 'Variable' and 'Description'.

Process data

Figure 1: Data upload tab. Users upload their dataset and can optionally provide a variable description file containing variable names and descriptions.

268 file is provided, the application will automatically use the variable names
269 themselves as default labels in all visualizations and summary tables.

270 In the **Variables** tab, they select the variables they wish to explore, op-
271 tionally assisted by a table of the variables' names and descriptions (see
272 Figure 2).

273 Next, users can adjust the association thresholds in the **Correlation Network**
274 tab to focus on the most meaningful relationships among the selected vari-
275 ables (see Figure 3).

276 Finally, the **Pairs Plots** tab displays detailed bivariate visualizations, in-
277 cluding scatter plots, mean plots, and contingency tables, for each retained
278 association (see Figures 4-6).

279 In addition, the **Help** tab provides a concise, step-by-step guide on how to
280 use the application (see Figure 7).

281 Each visualization panel includes contextual download controls to facilitate
282 the reuse of figures and tables (e.g., in reports or data-journalism pieces).

283 The correlation network can be exported directly as a PNG image, and each
284 pairwise association displayed in the Pairs Plots tab can likewise be down-

285 loaded as an image. Scatterplots, mean plots, and contingency tables are
286 saved with the same styling and labels as shown on screen, ensuring full vi-
287 sual consistency between the interactive interface and the exported outputs.

288 This example shows how non-expert users, such as journalists or engaged
289 citizens, can uncover unexpected or important relationships in a public opin-
290 ion dataset. These insights can serve as the starting point for local news,

Association Explorer

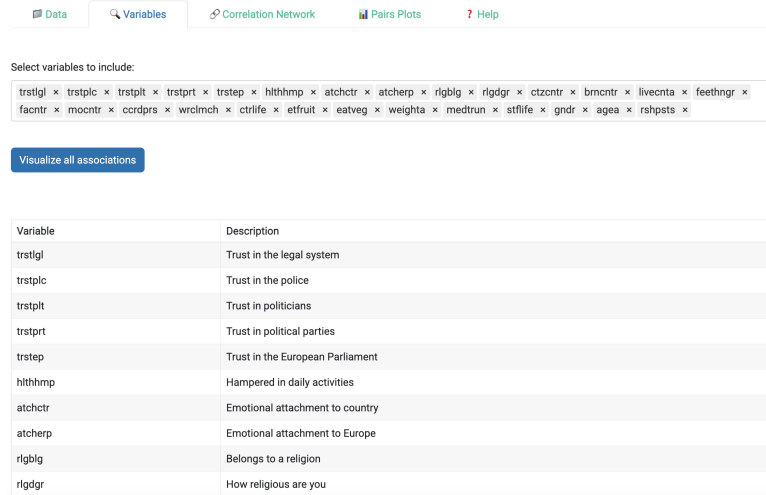


Figure 2: Variables tab. Users select the variables to include in the analysis, with access to descriptions (if provided in the data tab) for easier selection.

Association Explorer

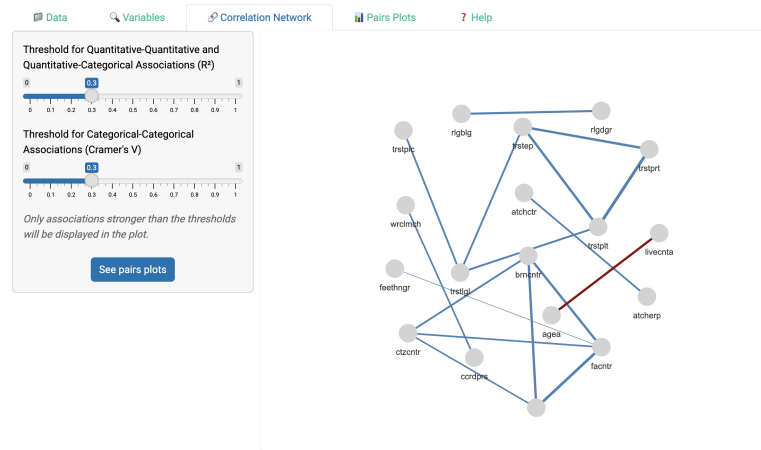


Figure 3: Interactive correlation network. Nodes represent variables and edges represent associations; edge thickness and length indicate strength, and edge color reflects direction (for numeric pairs only).

Association Explorer

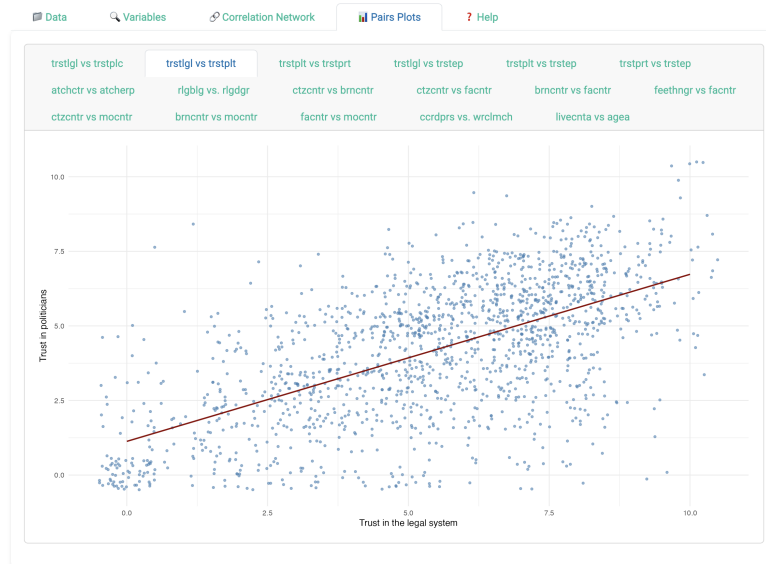


Figure 4: Example of a scatter plot for a numeric-numeric pair. The plot includes a linear regression line fitted via ordinary least squares (OLS).

Association Explorer



Figure 5: Example of a mean plot for a numeric-categorical pair. Categories are ordered by the mean value of the quantitative variable to aid interpretation.

Association Explorer

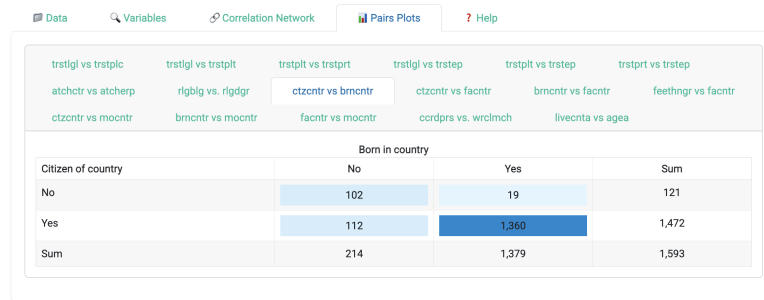


Figure 6: Example of a colored contingency table for a categorical-categorical pair, with color intensity reflecting frequencies and marginal sums displayed.

Association Explorer



How to use the Association Explorer app?

- Upload your dataset (CSV or Excel) in the 'Data' tab. Optionally, upload a file with variable descriptions. This file must contain 2 columns called 'Variable' and 'Description'.
- In the 'Variables' tab, select the variables you want to explore. If you upload a file containing variables' descriptions, a summary table below shows the selected variables along with their descriptions.
- Click 'Visualize all associations' to access the correlation network.
- Adjust the thresholds to filter associations by strength. Only variables that have strong associations (as defined by the thresholds) will appear in the network and pairs plots.
- In the network, thicker and shorter edges indicate stronger associations.
- Click 'See pairs plots' to display bivariate visualizations for retained associations.

Figure 7: Help tab, offering guidance on how to use the application.

291 public debates, or policy communication. A screencast demonstrating the
292 full workflow of the application is available in the documentation folder of
293 the GitHub repository.

294 Note that in this example, no survey weights have been applied. While the
295 ESS dataset includes weights to account for complex sampling designs, we
296 chose not to use them here in order to maintain clarity and responsiveness
297 in the interface, and because our focus is on demonstrating the app’s core
298 functionality rather than producing nationally representative statistics.

299 4. Impact

300 AssociationExplorer enables a broader range of users, including researchers,
301 students, journalists, and engaged citizens, to explore complex datasets in-
302 volving both quantitative and qualitative variables. By eliminating the need
303 for advanced programming or statistical knowledge, the software lowers the
304 barrier to entry for data-driven discovery and interpretation. The app facil-
305 itates the systematic screening of pairwise associations in survey or public
306 data, offering insights that can guide hypothesis generation, variable selec-
307 tion, or further multivariate modeling.

308 It also facilitates the assessment of relationships between variables of dif-
309 ferent types (numeric, ordinal, and nominal) within a single unified inter-
310 face. Unlike many tools that require preprocessing steps such as dummy
311 coding, AssociationExplorer handles categorical variables natively and auto-
312 matically selects the appropriate statistical association measure (i.e., Pear-
313 son’s r , Cramer’s V , correlation ratio η). This ensures that users can obtain
314 meaningful and interpretable results without needing to master complex pre-
315 processing steps or statistical diagnostics.

316 Beyond supporting research, we are convinced that automating these statisti-
317 cal processes can contribute to the promotion of open public data repositories.
318 In fact, the scientific literature points to a chronic underuse of open public
319 data [16], particularly due to the resources and knowledge required to pro-
320 cess these data. By lowering this barrier to entry, this tool makes it possible
321 to utilize databases that would otherwise remain inactive. The exploitation
322 and understanding of these data responds to crucial issues of transparency
323 in public action [19].

324 Although this tool is open to anyone, its development was particularly in-
325 spired by its use in a journalistic context. Data journalism, in particular, is
326 one of those examples of virtuous practice that we believe is best suited to
327 supporting the open data movement. This practice, which relies on the use
328 of digital data as material for journalistic investigation [6], is now used to
329 analyze election results, explain climate change, and track the evolution of

330 health crises. However, it comes at a cost, which explains why this practice
 331 is still limited to a few large national media outlets.
 332 However, it would be detrimental to leave these practices inaccessible to local
 333 newsrooms. By covering events that are often overlooked by national media,
 334 these local media outlets can highlight data that would otherwise be ignored
 335 (finance, consumption, police, municipal council records, etc.). Supporting
 336 data journalism at the local level can also help to enhance the value of this
 337 journalistic practice, which has been shown to strengthen social cohesion [13],
 338 stimulate civic participation [17], and influence public policies [10].
 339 While national media outlets are struggling, these challenges are even more
 340 pronounced for local media, which generally face the same problems but
 341 with even more limited resources [5]. As a result, few institutions have the
 342 means to meet the labor and skill requirements necessary for this investigative
 343 practice [4]. We hope that AssociationExplorer can help meet these needs.
 344 The app is also planned to be used as an interactive teaching aid in statistics
 345 and data literacy courses. It will allow students to upload real datasets and
 346 immediately visualize how variables relate, reinforcing theoretical concepts
 347 through intuitive, hands-on exploration. By presenting associations visually,
 348 the app encourages learners to engage with their data more actively and
 349 reflect critically on patterns, variable relationships, and data quality.
 350 Although already fully functional, the software is currently made available as
 351 a standalone open-source R Shiny application on GitHub and can be launched
 352 directly from R. At this stage, it is therefore only accessible to users who have
 353 R installed on their machine. As part of a broader research project called
 354 ODALON, which aims to build an integrated platform combining multiple
 355 tools for automated local news production, AssociationExplorer will serve
 356 as a key building block. A public, fully online version of the app will be
 357 released as the project progresses, and no later than December 2026 (which
 358 corresponds to the end of the project), ensuring that anyone, including those
 359 without access to R, can freely use the application through a user-friendly
 360 web interface.
 361 Future impact is expected through integration into platforms for civic data
 362 storytelling, local journalism, and public communication. While not intended
 363 for commercialization, the application’s role in supporting open, inclusive,
 364 and responsible data exploration reflects its potential for broader societal
 365 relevance.

366 **5. Conclusions**

367 AssociationExplorer provides a user-friendly, open-source solution for explor-
 368 ing statistical associations within multivariate datasets. By integrating ro-

369 bust statistical measures with dynamic and intuitive visualizations, the app
370 empowers non-expert users, such as journalists, educators, and engaged cit-
371 izens, to uncover patterns and relationships that might otherwise remain
372 hidden. Its ability to handle both quantitative and qualitative variables,
373 combined with an interactive interface and an easy-to-follow workflow, makes
374 it especially suitable for exploratory research, data storytelling, and public
375 communication.

376 Beyond its technical features, the application contributes to democratizing
377 access to statistical tools and supports data literacy in a wide range of con-
378 texts. The case study using the European Social Survey illustrates how users
379 can navigate from raw data to meaningful insights with minimal technical
380 knowledge.

381 While the app is currently accessible through R for users with basic techni-
382 cal setup, a fully web-based version will make it universally available as part
383 of a broader data exploration platform being developed under the ODALON
384 project. Before public deployment, performance will be systematically tested
385 on larger datasets to ensure responsiveness and reliability. Any identified lim-
386 itations, such as reduced interactivity with high-dimensional data or mem-
387 ory constraints, will be clearly communicated to users through the appli-
388 cation’s documentation. By lowering barriers to data exploration and in-
389 terpretation, AssociationExplorer supports more inclusive, transparent, and
390 evidence-informed engagement with complex social data.

391 Future improvements will be guided by user feedback and community con-
392 tributions. One planned feature is the ability to incorporate user-specified
393 weights, especially relevant for survey data. However, this functionality is
394 not yet implemented in the current version to preserve simplicity and ensure
395 responsiveness in the user interface, particularly for users unfamiliar with
396 survey weighting procedures. In the same spirit, future versions will also in-
397 clude a dedicated panel for basic graphic customization (e.g., color palettes,
398 titles, and labels). Although not implemented at the current stage to preserve
399 a lightweight and responsive interface for non-technical users, this feature is
400 planned for the forthcoming public web-based release to enhance usability
401 and flexibility.

402 Additional avenues for future research and development include extending
403 the application to compute partial correlations, which can help isolate direct
404 relationships between variables by controlling for confounders. Visualizations
405 involving three variables could also be introduced, for instance, scatter plots
406 where the size or color of points reflects a third variable, to uncover more nu-
407 anced patterns. Furthermore, users could be allowed to filter variables based
408 on missingness thresholds or variable types, and a module for longitudinal
409 data (e.g., panel structures or time series) could expand the app’s relevance

410 to temporal analyses. For contingency tables, adding conditional distribu-
411 tions (by row or by column) in addition to the joint distribution could provide
412 more informative insights. This would allow users to better explore subsets
413 of the data. For example, even if a group such is underrepresented in the
414 sample (and therefore lightly colored in the joint table), the conditional dis-
415 tribution would still reveal the relative patterns within that group. Another
416 enhancement could involve enabling two-sided intervals for the association
417 thresholds, rather than the current one-sided sliders. This would allow users
418 not only to focus on the strongest associations, but also to deliberately exam-
419 ine the weakest ones, thereby uncovering cases where variables appear largely
420 independent. In addition, users could be given the option to flip the X and
421 Y axes in quantitative–quantitative pairs plots, providing greater flexibility
422 when comparing variables and improving interpretability when relationships
423 are more intuitively viewed in the reverse orientation. Providing interpretive
424 text alongside tables and plots (especially if the application is deployed for
425 non-specialist audiences) could also help ensure that outputs are understood
426 correctly and used responsibly. Lastly, although the application includes
427 built-in export options for visualizations and tables, future extensions could
428 incorporate a dedicated panel for basic graphic customization, and even auto-
429 mated summaries of key associations, to further enhance its use in reporting
430 and collaborative workflows.

431 A more ambitious extension would integrate a large language model (LLM)
432 to enable users to enter a free-text prompt or a thematic query. The app
433 would then automatically identify variables that are semantically similar or
434 potentially associated with the theme, based on the dataset and the optional
435 description file provided by the user. These extracted variables would be
436 pre-selected in the interface, as if the user had chosen them manually, after
437 which the analysis would proceed as in the current version. This feature
438 could significantly enhance accessibility for users unfamiliar with the dataset
439 or its structure, and accelerate the discovery of relevant associations.

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